

FMUSER FUQ-4635B Encoder Modulator 1-4-8 HDMI Inputs MPEG2 Encoding Multi-Standard RF DVB-C-DVB-T-ATSC-ISDB-T-DTMB Web NMS Control Broadcast Systems Cable TV Headend Product Overview



The FUQ-4635D Encoder Modulator integrates MPEG2 encoding and modulation, supporting scalable 1/4/8 HDMI inputs and multi-standard RF outputs including DVB-C, DVB-T, ATSC, ISDB-T, and DTMB. It converts audio/video signals into encoded and modulated RF formats. This device features a modular design and web-based NMS for control. Ideal for broadcast systems and cable TV headends.

Key feature

Supports scalable 1/4/8 HDMI inputs for flexible source integration.

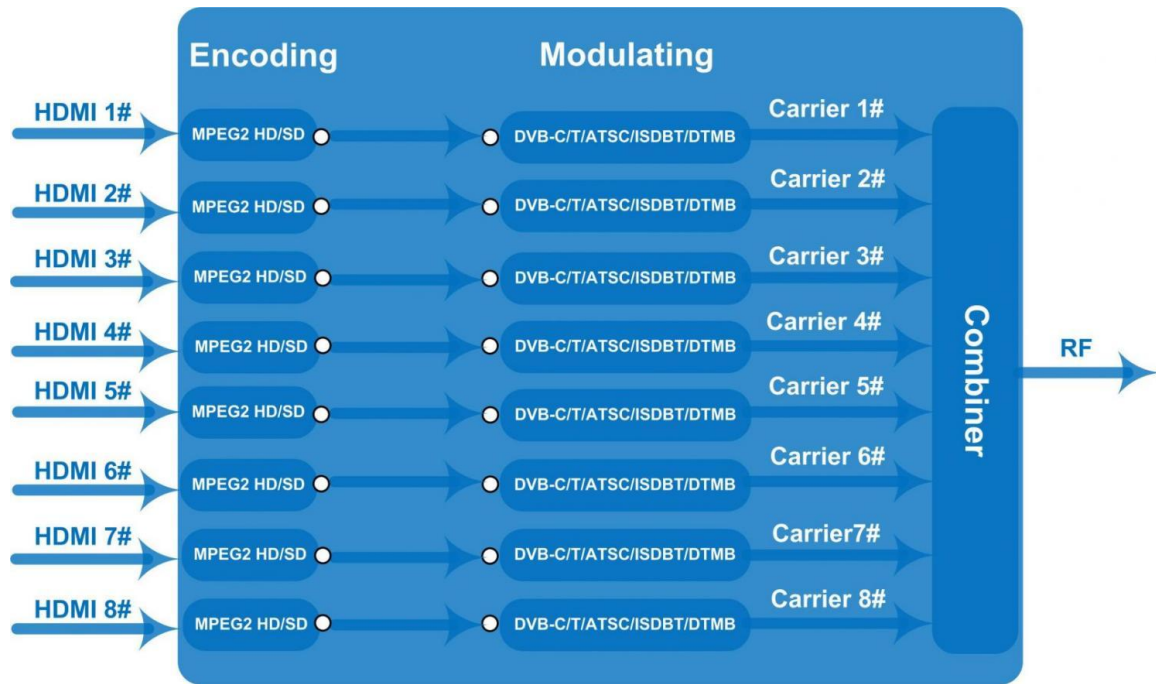
Features MPEG2 video encoding with resolutions up to 1080p/i, 720p, 576i, and 480i.

Offers low latency operation from 50ms to 1000ms.

Provides multi-standard RF output including DVB-C, DVB-T, ATSC, ISDB-T, and DTMB with dynamic switching.

Includes web-based NMS for easy control and updates.

Inner Principle Chart



Specifications

HDMI Encoding Section	HDMI Encoding Section	HDMI Encoding Section	HDMI Encoding Section
Input	Input	Interface	HDMI*1/4/8
Video	Video	Encoding	MPEG2
Video	Video	Input	Output
Video	Video	480@59.94/60i	480@30p
Video	Video	576@50i	576@25p
Video	Video	720@50/59.94/60p	720@25/30p
Video	Video	1080@50i	1080@25p
Video	Video	1080@59.94/60i	1080@30p
Video	Video	1080@50p	1080@25p
Video	Video	1080@59.94/60p	1080@30p
Video	Video	Bit-rate	2.000~20.000 Mbps
Video	Video	GOP	1~60
Video	Video	Rate Control	VBR
Video	Video	Aspect Ratio	16 :9
Audio	Audio	Encoding	MPEG1 layer 2,MPEG-2 AAC, AC3

Audio	Audio	Sample rate	48KHz
Audio	Audio	Bit rate	MPEG1 layer 2 (128kbps)
Audio	Audio	Bit rate	MPEG-2 AAC (VBR)
Audio	Audio	Bit rate	AC3 (128kbps)
DVB-C Modulator Section	DVB-C Modulator Section	DVB-C Modulator Section	DVB-C Modulator Section
Standard	Standard	J.83A (DVB-C), J.83B	J.83A (DVB-C), J.83B
MER	MER	≥35dB	≥35dB
RF frequency	RF frequency	100-1000MHz, 1KHz step	100-1000MHz, 1KHz step
RF output level	RF output level	-48~ -10dBm, 1dB step	-48~ -10dBm, 1dB step
Symbol rate	Symbol rate	3.000~7.000Mpsps adjustable	3.000~7.000Mpsps adjustable
		J.83A	J.83B
Constellation	Constellation	16/32/64/128/256QAM	64/ 256QAM
Bandwidth	Bandwidth	8M	6M
DVB-T Modulator Section	DVB-T Modulator Section	DVB-T Modulator Section	DVB-T Modulator Section
Standard	Standard	DVB-T COFDM	DVB-T COFDM
Bandwidth	Bandwidth	2M, 3M, 4M, 5M, 6M, 7M, 8M	2M, 3M, 4M, 5M, 6M, 7M, 8M
Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Code rate	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Guard Interval	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Transmission Mode	Transmission Mode	2K, 8K	2K, 8K
MER	MER	≥35dB	≥35dB
RF frequency	RF frequency	100-1000MHz, 1KHz step	100-1000MHz, 1KHz step
RF output level	RF output level	-48~ -10dBm, 1dB step	-48~ -10dBm, 1dB step
ISDB Modulator Section	ISDB Modulator Section	ISDB Modulator Section	ISDB Modulator Section

Standard	Standard	ARIB STD-B31	ARIB STD-B31
Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Guard Interval	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Transmission Mode	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K
Code rate	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
RF frequency	RF frequency	100~1000MHz, 1KHz step	100~1000MHz, 1KHz step
RF output level	RF output level	-48~ -10dBm, 1dB step	-48~ -10dBm, 1dB step
ATSC Modulator Section	ATSC Modulator Section	ATSC Modulator Section	ATSC Modulator Section
Standard	Standard	ATSC A/53	ATSC A/53
Constellation	Constellation	8 VSB	8 VSB
RF output level	RF output level	-48~ -10dBm, 1dB step	-48~ -10dBm, 1dB step
MER	MER	≥35dB	≥35dB
RF frequency	RF frequency	100~1000MHz, 1KHz step	100~1000MHz, 1KHz step
DTMB Modulator Section	DTMB Modulator Section	DTMB Modulator Section	DTMB Modulator Section
RF frequency	RF frequency	100-1000MHz, 1KHz step	100-1000MHz, 1KHz step
RF output level	RF output level	-48~ -10dBm, 1dB step	-48~ -10dBm, 1dB step
Transmission Mode	Transmission Mode	S-0.8-4QAM-595, S-0.8-16QAM-595, M-0.4-16QAM-945V, M-0.6-16QAM-945V, M-0.6-64QAM-420V, M-0.6-64QAM-945V, M-0.8-16QAM-420V, M-0.8-16QAM-945V, M-0.8-64QAM-420V,	S-0.8-4QAM-595, S-0.8-16QAM-595, M-0.4-16QAM-945V, M-0.6-16QAM-945V, M-0.6-64QAM-420V, M-0.6-64QAM-945V, M-0.8-16QAM-420V, M-0.8-16QAM-945V, M-0.8-64QAM-420V,
System	System	System	System
Management	Web	Web	Web
Language	English	English	English

Upgrade	Web update	Web update	Web update
General	General	General	General
Power supply	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz
Dimensions	482*328*44.5mm	482*328*44.5mm	482*328*44.5mm
Temperature	0~45■(operation), -20~80■(storage)	0~45■(operation), -20~80■(storage)	0~45■(operation), -20~80■(storage)

Technical Comparison

	FU-4653A	FU-4653I	FU-4635C
Chipset	Vixs	Hisilicon	VATek
Stream Input	1*ASI input for re-mux	1 DVB-C tuner	N/A
IP Input	N/A	128*IP	N/A
Video Channel	4* HDMI/SDI input optional	4/8/12*HDMI input optional	1/4/8*HDMI input optional
Video Encode	MPEG2 & MPEG4 AVC/H.264	MPEG-4 AVC/H.264	MPEG2
Video Bit Rate	0.8~19Mbps for H.264 encoding 1~19Mbps for MPEG-2 encoding	1Mbps~13Mbps each channel (Bit rate is with a high fluctuation, suitable for cable system, not suitable for satellite uplink)	2~20Mbps each channel
Resolution	Up to 1920*1080@50P/60P (MPEG4 AVC/H.264) Up to 1920*1080@50I/60I (MPEG2 HD)	Up to 1920*1080@50P/60P	Up to 1920*1080@50P/60P
Audio Encode	MPEG1 Layer 2, MPEG2-AAC, MPEG4-AAC, AC3 2.0 (Optional)	MPEG-1 Layer 2	MPEG1 layer 2, MPEG2-AAC, AC3
Audio Gain	N/A	Yes	N/A
OSD Function	N/A	Yes	N/A

CC Input	CC (Closed Caption)for SDI interface (Optional), EIA608, EIA708	N/A	CC (Closed Caption), EIA 608
Low Latency	Yes	N/A	Yes
ASI output	Yes	N/A	N/A
IP output	4 SPTS or 1 MPTS output	4 MPTS IP output	N/A
Modulating	4*DVB-C/2* DVB-T/4*ATSC-T /1/2*ISDB-T RF out	4*DVB-C/DVB-T RF out	4/8*DVB-C/T/ATSC/IS DB-T RF out in one device, Modulation dynamic switching
RF Frequency	30~960MHz, 1KHz step	50~960MHz, 1KHz step	100-1000MHz, 1KHz step

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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